

AGATA, Russian Federation

WMO: 233830

Lat: 66.926N

Lon: 93.493E

Elev: 278

StdP: 98.03

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-51.8	-49.5	-53.5	0.0	-52.0	-51.4	0.0	-49.4	8.3	-18.4	7.4	-16.7	0.0	30	0.816

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	25.2	16.3	22.4	14.6	20.1	13.6	17.0	23.0	15.8	20.8	14.5	19.0	2.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.9	18.6	13.6	10.0	17.5	12.3	9.2	16.6	48.9	23.4	45.3	21.0	41.7	18.9	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	6.8	5.9	DB	-52.0	28.2	2.6	2.4	-53.9	29.9	-55.4	31.4	-56.8	32.7	-58.7	34.5	(4)
(5)			WB	-51.9	19.2	2.8	1.3	-53.9	20.2	-55.5	20.9	-57.1	21.7	-59.1	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.0	-34.5	-31.4	-19.6	-10.0	0.0	9.2	14.2	10.3	3.5	-6.1	-24.3	-32.6	(6)
(7)		DBStd	18.73	9.92	9.03	9.42	7.22	5.10	4.36	3.34	3.33	4.13	6.84	11.08	10.53	(7)
(8)		HDD10.0	7522	1381	1160	917	599	310	67	4	37	198	501	1028	1319	(8)
(9)		HDD18.3	10343	1639	1393	1176	849	568	274	135	249	446	759	1278	1577	(9)
(10)		CDD10.0	227	0	0	0	0	1	44	133	47	2	0	0	0	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	6	1	0	0	0	0	(11)
(12)		CDH23.3	151	0	0	0	0	1	21	118	11	0	0	0	0	(12)
(13)		CDH26.7	27	0	0	0	0	0	3	24	1	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	0.6	0.7	1.5	2.3	2.2	1.7	1.7	1.9	2.3	2.5	1.1	0.5	(14)
(15)	Precipitation	PrecAvg	464	26	25	19	21	28	55	60	67	54	45	36	31	(15)
(16)		PrecMax	639	60	264	44	68	61	119	172	113	130	90	73	60	(16)
(17)		PrecMin	261	7	3	2	2	1	2	1	9	16	14	5	18	(17)
(18)		PrecStd	95	13	41	10	14	15	28	34	28	23	18	15	10	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.5	-6.1	1.8	7.5	17.0	26.6	29.3	25.0	15.5	8.6	1.6	-5.6	(19)
(20)			MCWB	-7.6	-7.2	-0.9	2.9	9.1	16.1	17.3	17.5	10.9	6.0	0.2	-6.6	(20)
(21)		2%	DB	-11.2	-9.7	-0.8	5.1	12.1	22.9	26.5	21.0	12.8	5.5	-1.3	-9.2	(21)
(22)			MCWB	-12.0	-10.6	-2.6	1.7	6.6	14.1	16.5	14.1	9.0	4.0	-2.4	-10.0	(22)
(23)		5%	DB	-15.4	-13.7	-3.4	3.2	9.4	20.4	24.1	18.6	11.0	3.4	-4.8	-13.3	(23)
(24)			MCWB	-16.0	-14.5	-4.9	1.1	4.8	12.3	15.9	13.5	8.1	2.0	-5.7	-13.8	(24)
(25)		10%	DB	-19.2	-17.5	-5.7	1.1	7.0	17.9	21.7	16.5	9.4	2.0	-8.8	-16.6	(25)
(26)			MCWB	-19.5	-18.0	-6.9	-0.7	3.3	11.4	14.8	12.4	7.0	0.7	-9.7	-17.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.4	-7.2	-0.8	3.6	9.4	17.2	18.9	17.4	12.7	6.2	0.2	-6.1	(27)
(28)			MCDWB	-6.4	-6.1	1.1	6.4	15.8	25.0	25.3	23.1	14.9	8.1	1.6	-5.2	(28)
(29)		2%	WB	-12.0	-10.6	-2.7	2.2	7.0	15.0	17.6	15.3	9.9	3.9	-2.5	-10.1	(29)
(30)			MCDWB	-11.2	-9.8	-0.9	4.3	11.5	20.6	23.6	19.2	11.3	5.3	-1.4	-9.3	(30)
(31)		5%	WB	-16.0	-14.4	-4.8	1.1	5.2	13.4	16.7	14.0	8.6	2.2	-5.6	-13.9	(31)
(32)			MCDWB	-15.3	-13.7	-3.4	3.3	8.6	18.6	22.3	17.6	10.5	3.2	-4.7	-13.4	(32)
(33)		10%	WB	-19.6	-18.0	-7.0	-0.8	3.7	11.9	15.7	12.9	7.0	0.7	-9.6	-17.0	(33)
(34)			MCDWB	-19.2	-17.5	-5.6	1.2	6.6	17.1	20.7	15.9	9.2	2.0	-8.9	-16.6	(34)
(35)	Mean Daily Temperature Range		MDBR	7.8	9.5	12.3	12.2	8.5	10.1	10.5	8.7	6.4	5.0	7.8	7.9	(35)
(36)		5% DB	MCDBR	9.8	10.8	9.0	9.8	11.6	16.4	16.8	14.3	9.9	4.3	5.8	9.5	(36)
(37)			MCWBR	9.5	10.3	8.0	7.6	7.0	8.9	8.2	8.3	6.8	3.8	5.4	8.9	(37)
(38)		5% WB	MCDWB	9.9	10.8	8.8	9.0	10.5	14.0	13.4	11.5	8.0	3.9	5.7	9.5	(38)
(39)			MCWBR	9.5	10.3	7.8	7.2	6.7	8.5	7.3	7.2	6.1	3.6	5.5	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.209	0.257	0.303	0.330	0.349	0.366	0.414	0.401	0.324	0.255	0.191	N/A	(40)	
(41)		taud	1.842	1.982	2.004	2.027	2.101	2.355	2.247	2.236	2.371	2.093	1.877	N/A	(41)	
(42)		Ebn at Noon	136	546	718	809	838	838	773	736	725	549	147	0	(42)	
(43)		Edh at Noon	15	58	98	125	130	105	114	104	70	49	13	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.05	0.62	2.17	4.34	5.83	5.52	4.82	2.99	1.69	0.65	0.13	0.00	(44)	
(45)		RadStd	0.01	0.05	0.20	0.34	0.41	0.60	0.57	0.34	0.28	0.12	0.02	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page